

COURSE NAME:

SIMATS ENGINEERING ASSESSMENT TOOL 1:
Short Answer Test

Cloud Computing and COURSE
CODE: Big Data Analytics

CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)

Question Count: 5

Total Marks: 25

Code of Conduct

I Logeshwaran R(1925111s) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Loggle

Assessment Tasks

Short Answer Test

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1. Define cloud computing and explain any four characteristics that make it different **from** traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show **how these** changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate **between** virtualization **as** an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable **real-world** example for each model.

Short Answer Test Rubric

		Level 4- Exemplary Accurate
Criteria	Weightage	
Conceptual Accuracy	30%	definitions and precise explanation of all key terms
Coverage of Concepts	20%	Covers all expected sub-points completely
Use of Examples	15%	Uses highly
Comparison / Differentiation	20%	

Clarity of

15%

Expression

relevant cloud
examples to
strengthen the
answer

Clearly
distinguishes
concepts with strong
logic Clear.
organized and
concise answer
writing

Level 3- **Proficient**

Mostly accurate with
minor gaps

Covers most
expected points

Uses relevant
examples with minor
mismatch

Reasonable
differentiation
with minor overlap
Generally clear
with few issues

Level 2- **Developing** Partial understanding

with noticeable
errors

Covers only
basic points

Uses vague or
partially
relevant
examples

Comparison is
weak or incomplete

Understandable but
poorly organized

Level 1 -
Beginning
Incorrect or irrelevant
explanation

Very limited
coverage

No useful
examples

Fails to differentiate

concepts

Difficult to
follow

1. Definition

USA 1506

Cloud completing and Big
Dada

SLOT D.

cloud computing is the delivery of
computing

storage, databases, networking

services such as

servers

and

software

software over

over the

Internet

m

a pay-

ал

you use

ин

basis.

Characteristics:

* on-demand self service users can access computing resources whenever needed without human

intervention

* Broad network access

services are available

over the Internet and can be accessed from computers laptops and mobile devices.

* Resource pooling - Resources are shared among multiple users using virtualization for efficient utilization

* Rapid elasticity

Resources can be quickly

increased, decreased based on demand.

Difference

.

дится

(loud provides
scalable

2. Evolution:

where

internet band, pay prouse

as on premise requires purchging
hardwan locally

Hard ware evolution:

Main frame computer →

Personnel

computer → High performance
seruets.

Internet

* Network evolution.

LAN -> WAN ~

High

speed

Multiple

virtual machines

* virtualization:

run

on one physical server,
improving

utilization.

* web technologies

improving resource

web applications and web
software access through browser.

services enabled software
access

* cloud platform **from**: combining
virtualization, high speed internet, and distributed

Conclusion:

data.

These advancement made cloud computing

flexible and acessible.

scalable,

3. Role of server virtualization:

Server virtualization

allows multiple

vistual

machines

to

run on

a single physical

server using

a

utilization

hypervisor. This improves hard

wan

enables quick development.

Virtualization

*Technology that creates

virtud machine

* Forum on efficient

hardware utilization.

*

works within a server

* EX. VM ware, Hyperv

cloud computing

*first model that deliver

IT resource

over Internet.

* Fours on provide

servizy

like

IaaS, PaaS, & SaaS

* are virtualization

EX. AWS EC2, Azure

conclusion:

virtualization

cloud computing, while cloud computing
delivers those

is the foundation that
enables

virtualized resources

as

services.

4. Definition:

contains

A data center is a facility that
serves, storage devices, networking equipment
and power backup to

support cloud
services.

Poles of Data Centres:

- * Store and process large amount of data. *
- Host cloud applications and virtual machine. *
- Ensure high availability through redundancy and backup

control.

- * Provide security using firewalls and access

- * Ensure disaster recovery

continuity

y

and business

* support scalability by adding resources

when demand

Example:

instances

Google cloud and
AWS.

5.

Model

* IaaS

* PaaS

Description

Provides virtual servers,

Storage and

networking.

User
manage os and
application

Provides a platform to
develop, test and
deploy

Example

Amazon EC2

Google
app
engine

applications without
managing hardware

* SaaS

Provides ready to use

Gmail

software over
the
internet

★ XaaS

Any IT services
delivered

Dropbox.

through the cloud, such as
storage, databass

conclusion:

IaaS provide infrastrucutun, PaaS provides
development **platform**, SaaS provides complete
softuen,
includes all othe cloud based service.

and KaaS